7 Limitations

The task is closed-vocabulary, short-answer, and scored by exact match. This gives the clean binary correctness signal a calibration study needs, and the secret-word mechanism is the same one used to elicit hidden objectives in the auditing work we cite, but realistic oracle outputs contain multiple claims, hedges, and explanations. Whether these results transfer to claim-level or sequence-level calibration of open-ended oracle answers is untested; the constrained-label result is a first step toward it, and we consider full claim-level calibration the main open problem.

All four oracles descend from one training recipe lineage (LatentQA with classification and context-prediction data). The ranking's stability across family and scale is evidence of generality across models, and it says nothing about oracles trained with substantially different recipes, such as the conversational-data recipe of Bauer et al. (2026).

One confound does not apply here. Jakkli et al. (2026) caution that an oracle can succeed by reconstructing the text near the activations it is given, without reading any internal state. The secret-word task rules that out by construction, because the target models are trained never to say their word, so the correct answer never appears in the surface text the activations come from. Oracle tasks whose answer is present in the target's text carry the confound, and our results say nothing about those.

The answer-extraction pipeline is shared across methods, so the method ranking is invariant to it; absolute accuracy would shift under an embedding-based soft metric that credits near-synonyms. Forced choice requires an enumerable candidate set, and its numbers should not be compared to the free-text rows as if it solved the same problem.

All prompts, secret words, and oracle outputs are English; calibration under other languages is unmeasured.

8 Ethical Considerations

No human-subjects data is used. The secret words and context prompts are reused from Karvonen et al. (2025) and contain no personal information; the 20 target models are synthetic research artifacts.

Calibrated confidence makes activation oracles more usable in audit pipelines, and that cuts both ways. The benefit is fewer wrong actions: a calibrated auditor discards low-confidence recoveries instead of acting on them. The residual risk is the confident wrong answer: on this task the expensive failure mode is a topical neighbour delivered with high confidence, and a pipeline that trusts calibrated scores will still act on those. A second risk is dual use: the same recovery-plus-confidence machinery that audits a model for hidden objectives could be pointed at models fine-tuned on private data. Our task uses synthetic secrets precisely to study the mechanism without that exposure, and the released artifacts are intended for research use consistent with the licences of the models they derive from.

Compute and energy are reported in section N.

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A Full scorecard

Tables 3 and 4 report every method-temperature row with all five metrics, for the Qwen pair and the Gemma pair respectively.

B Bootstrap 95% CIs

Table 5 reports 1000-resample bootstrap percentile intervals for the main scorecard rows on all four oracles, resampling the n=6,000 per-sample (confidence, correctness) pairs. Standard errors on AUROC are around 0.003 to 0.008.

C Post-hoc calibration

Tables 6 through 9 report test-set ECE after fitting four post-hoc calibrators on a held-out slice, for each oracle. Calibrators: *Temperature* (one-parameter logit rescale, Guo et al., 2017), *Platt* (two-parameter logistic regression on $\mathrm{logit}(p)$, Platt, 1999), *Isotonic* (non-parametric monotone, Zadrozny and Elkan, 2002), and *Beta* (logistic regression on $[\log p, -\log(1-p)]$, Kull et al., 2017). *Word-disjoint* fits on a fixed random 10 of the 20 secret words (seed 1) and tests on the other 10; *Random 50/50* is a fixed sample-level split (seed 2). The same split is used for every method within an oracle, so method comparisons are not confounded by split randomness.

All four calibrators close most of the absolute ECE gap on the random split, including for the two weakest scores (direct, raw MCMC acceptance), because the calibrator acts as a label-trained classifier on top of the score; AUROC is monotone-invariant and is what still separates methods. The word-disjoint split stays substantially harder, and on Qwen3-8B the isotonic fit degrades the already calibrated bootstrap T=1.0 (ECE 0.061 $\rightarrow$ 0.094), a reminder that a calibrator fit under word shift can hurt.

D Bootstrap sample-count ablation

Table 10 recomputes the bootstrap confidence from the first k of the 20 saved draws, at each oracle's ECE-optimal temperature from the original grid.

E Calibration and ranking diagnostics

Figure 2 plots each method on the calibration-versus-ranking plane for all four oracles.

Figure 3 shows reliability diagrams (binned mean confidence versus accuracy) for four selected methods on all four oracles. The bootstrap at the tuned temperature tracks the diagonal; log-probability is underconfident on the Qwen oracles; direct numeric self-report collapses to a single high-confidence bin whose accuracy equals the overall task rate.

F Per-word accuracy

Table 11 reports per-word accuracy on bootstrap T=1.0 for all four oracles.

G Controlled target-set scaling

Table 12 reports the controlled-N variant on the two Qwen oracles, restricting the target vocabulary to $N \in \{2, 5, 10, 20\}$ words with a fixed seed: N=2: {snow, ship}; N=5: {flame, green, leaf, rock, snow}; N=10: {gold, song, chair, wave, ship, blue, snow, smile, flame, flag}; N=20: the full vocabulary. The samples per row are $300 \cdot N$ for $N \leq 10$ and 6,000 for N=20.

H Per-layer readout sweep

Each verbalizer is trained on activations from three layers (the 25%, 50%, 75% depth points of Karvonen et al. (2025)). At inference time the experiment reads from one of these three. We swept every layer index $0..N-1$ on four oracle bases: Qwen3-8B, Qwen3.6-27B, Gemma-2-9B, and Gemma-3-27B. The Qwen3-8B and Gemma-2-9B verbalizers are released by Karvonen et al. (2025), that demonstrate the secret-word taboo task on those two models. The Gemma-3-27B verbalizer was trained using the same SFT recipe. Under the active target LoRA we captured the residual stream, fed it to the verbalizer with the matching introspection prefix, and recorded log-prob baseline accuracy across 20 words $\times$ 5 contexts (n=100 per layer per model). Trained layer indices: $\{9, 18, 27\}$ on Qwen3-8B; $\{16, 32, 48\}$ on Qwen3.6-27B; $\{10, 21, 31\}$ on Gemma-2-9B; $\{15, 31, 46\}$ on Gemma-3-27B.

On every oracle the probe-readable region is a band several layers wide, but its position along the stack varies across the four bases (fig. 4). Qwen3-8B has a wide plateau of 35–50% across layers

Method	Qwen3-8B Acc	ECE	Brier	NLL	AUROC	Qwen3.6-27B Acc	ECE	Brier	NLL	AUROC
Forced choice (M8)	**0.646**	0.078	**0.100**	**0.324**	**0.940**	**0.533**	0.080	**0.100**	**0.332**	**0.936**
Log-prob (no offset)	0.414	0.256	0.249	0.748	0.824	0.234	0.131	0.157	0.484	0.842
Log-prob (with offset)	0.414	0.255	0.246	0.730	0.840	0.234	0.131	0.158	0.489	0.835
Bootstrap T=0.3	0.419	0.334	0.303	2.636	0.784	0.233	0.479	0.369	2.367	0.837
Bootstrap T=0.5	0.424	0.193	0.213	1.029	0.810	0.231	0.327	0.222	0.800	0.864
Bootstrap T=0.7	0.414	0.097	0.171	0.558	0.830	0.230	0.218	0.157	0.514	0.863
Bootstrap T=1.0	0.402	**0.057**	0.163	0.498	0.829	0.225	0.147	0.130	0.431	0.851
Bootstrap T=1.3	0.387	0.076	0.171	0.519	0.823	0.209	0.125	0.125	0.415	0.837
Bootstrap T=1.5	0.367	0.083	0.173	0.522	0.824	0.206	0.103	0.129	0.422	0.812
Bootstrap T=1.75	–	–	–	–	–	0.188	0.097	0.125	0.412	0.801
Bootstrap T=2.0	–	–	–	–	–	0.172	0.077	0.122	0.404	0.770
Constrained label (expected)	0.414	0.440	0.423	1.152	0.819	0.232	0.329	0.259	0.713	0.819
Constrained label $P(\text{high}+)$	0.414	0.548	0.530	1.946	0.819	0.232	0.256	0.212	0.612	0.801
Constrained label $P(\text{very high})$	0.414	0.425	0.387	1.064	0.812	0.232	**0.069**	0.158	0.478	0.778
MCMC agreement T=0.125	0.415	0.493	0.461	7.823	0.669	0.236	0.668	0.620	10.87	0.653
MCMC agreement T=0.25	0.418	0.408	0.371	4.673	0.737	0.233	0.569	0.482	5.658	0.767
MCMC agreement T=0.5	0.413	0.263	0.247	1.608	0.803	0.228	0.390	0.274	1.383	0.858
Steering sensitivity	0.418	0.404	0.354	4.401	0.763	0.232	0.501	0.413	4.172	0.725
MCMC accept T=0.125	0.415	0.544	0.534	10.77	0.563	0.234	0.740	0.729	15.61	0.528
MCMC accept T=0.25	0.403	0.547	0.532	10.28	0.579	0.230	0.738	0.722	15.04	0.547
MCMC accept T=0.5	0.376	0.551	0.531	9.358	0.601	0.210	0.746	0.722	13.97	0.574
Direct (numeric)	0.414	0.582	0.580	12.66	0.516	0.234	0.753	0.752	16.02	0.404

Table 3: Full method scorecard on Qwen3-8B and Qwen3.6-27B, n=6,000 samples per row. ECE, Brier, and NLL: lower is better. Accuracy and AUROC: higher is better. Best per column in **bold**. '–': configuration not run on this oracle (the extended temperatures were run only where the ECE optimum was not interior to the original grid).

9–32 peaking at layers 22–30; Qwen3.6-27B has a tighter band of 30–38% across layers $\sim$ 37–58 ($\sim$ 58–91% depth), with everything before layer $\sim$ 28 at the overall task rate. Gemma-2-9B reads early: its band sits between $\sim$ 24% and $\sim$ 50% depth and decays through the second half of the stack. Gemma-3-27B has the widest band of the four, $\sim$ 50–56% across layers 31–59 ($\sim$ 50–95% depth), with monotone rise from layer 11 and a sharp drop at the head. The gap between layers 32 and 48 on Qwen3.6-27B reported in section 4 is the distance between the edge and the inside of a single readout band; both layers lie inside the probe-readable region.

The trained sample triple $\{25, 50, 75\}$% is sparse with respect to these bands. On Qwen3.6-27B only the 75% sample lands inside the band; on Gemma-2-9B only 25% lands; on Qwen3-8B both the 50% and 75% samples are inside; and on Gemma-3-27B both the 50% and 75% samples land squarely in the plateau. Trained-vs-peak gaps at the best trained sample are tight on all four: 3 pp on Qwen3-8B (layer 27 at 47% vs. peak 50%), 3 pp on Qwen3.6-27B (layer 48 at 35% vs. layer 51 at 38%), 5 pp on Gemma-2-9B (layer 10 at 28% vs. layer 13 at 33%), and 3 pp on Gemma-3-27B (layer 46 at 53% vs. layer 40 at 56%).

I Attention backend numerics

The 27B run is sensitive to the choice of attention backend. With `flash_attention_4` on Qwen3.6-27B at sequence length 171 (roughly the length of one verbalizer plus one context prompt), the cosine similarity between the read-layer hidden state and the `flash_attention_2` reference drops to the 0.87–0.97 range, and the maximum absolute logit difference reaches 9.66. `eager` and `sdpa` remain numerically faithful at cos. sim. $>$ 0.9999. The regression surfaces only on the non-cached forward pass that the experiment uses for activation collection; greedy decoding through `model.generate` produces token sequences byte-identical to the reference under `flash_attention_4`. The same comparison on Qwen3-8B at similar sequence lengths shows agreement at the level of bfloat16 numerical noise across all four backends. All experiments in this paper use `flash_attention_2`.

J Methods designed for the steered setting

Alongside the five methods of section 3.3, we designed three confidence readouts that use the steering machinery itself. This appendix gives the background each one draws on, its procedure, and its result. Table 13 reports all three on all four oracles. Every row there is weaker on both calibration and ranking than the tuned bootstrap of table 1. We report them in full because the reason each one